

WELCOME

CCBP 4.0

21st Century Skills

Nested Loops

Daily Challenge

Identify The Mistake

Code

```
n = int(input())
k = n
for i in range(1, n):
    spaces = " " * k
    stars = "* " * i
    print(spaces+stars)
    k = k - 1
```

Input

```
3
```

Expected Output

```
  *
 * *
* * *
```

Output

```
  *
 * *
```

NXT WAVE

Daily Challenge

Identify The Mistake

Code

```
n = int(input())
k = n
for i in range(1, n+1):
    spaces = " " * k
    stars = "* " * i
    print(spaces+stars)
    k = k - 1
```

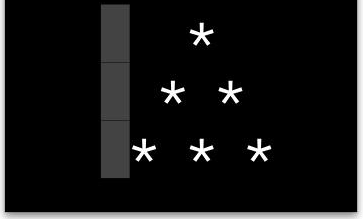
Input

```
3
```

Expected Output

```
  *
 * *
* * *
```

Output



NXT WAVE

Daily Challenge

Identify The Mistake

Code

```
n = int(input())
k = n - 1
for i in range(1, n+1):
    spaces = " " * k
    stars = "* " * i
    print(spaces+stars)
    k = k - 1
```

Input

```
3
```

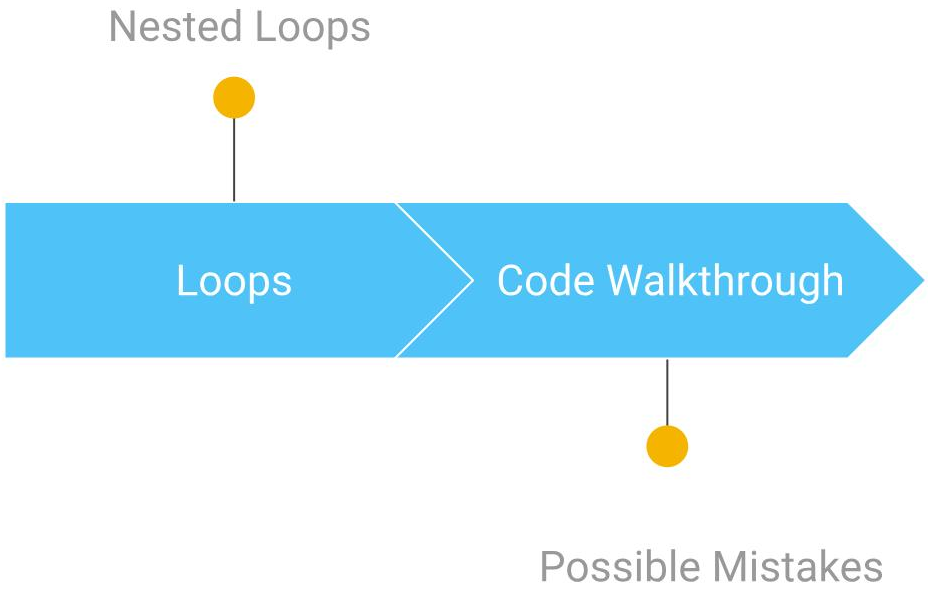
Output

```
  *
 * *
* * *
```

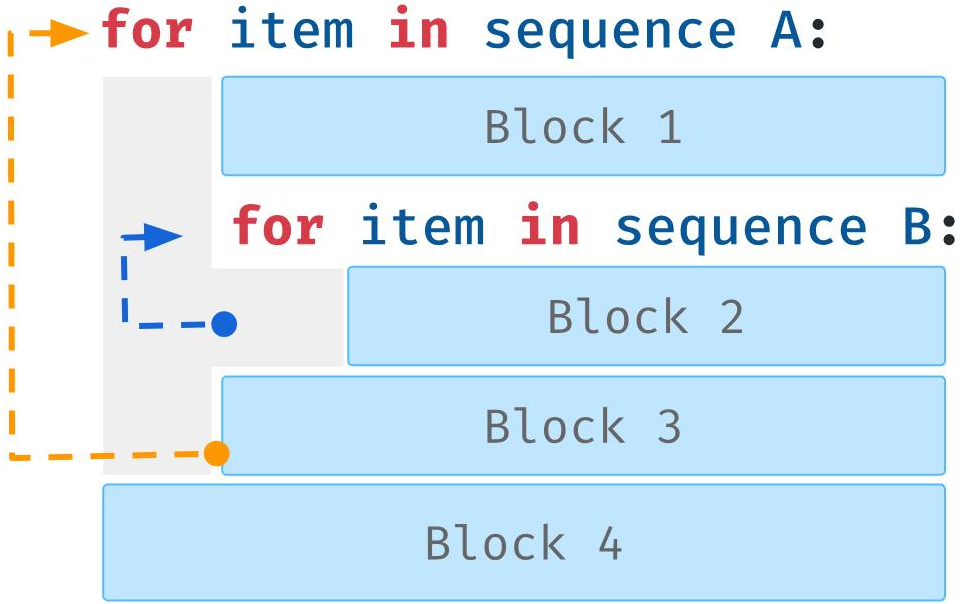
NXT WAVE

Agenda for Today's Session

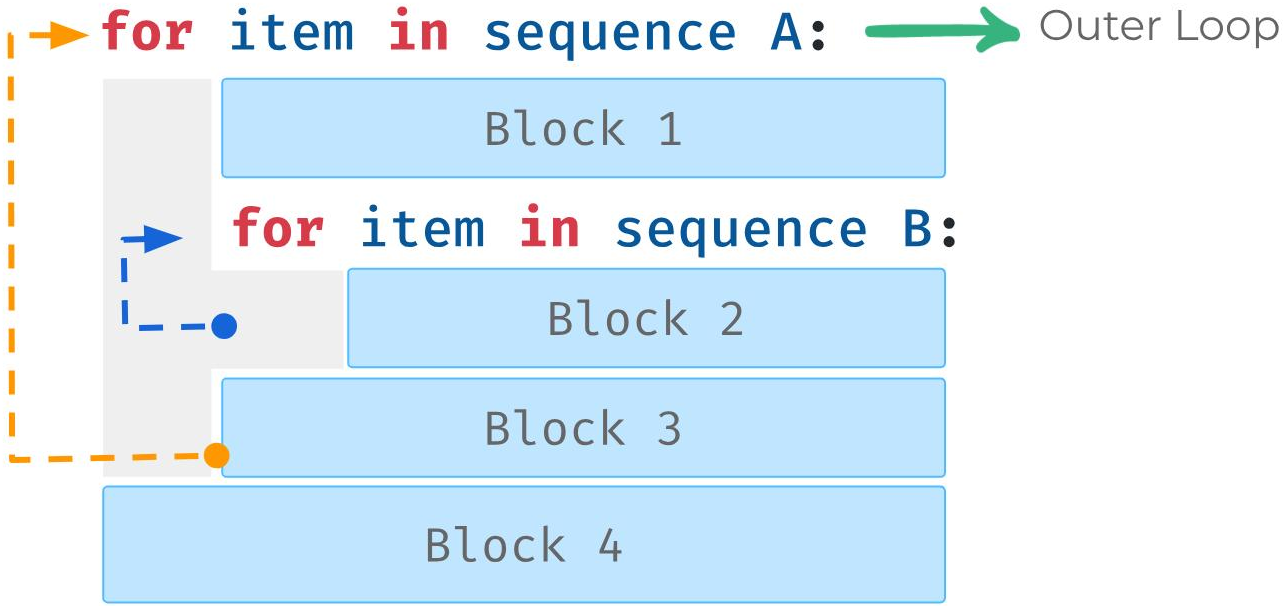
Nested Loops



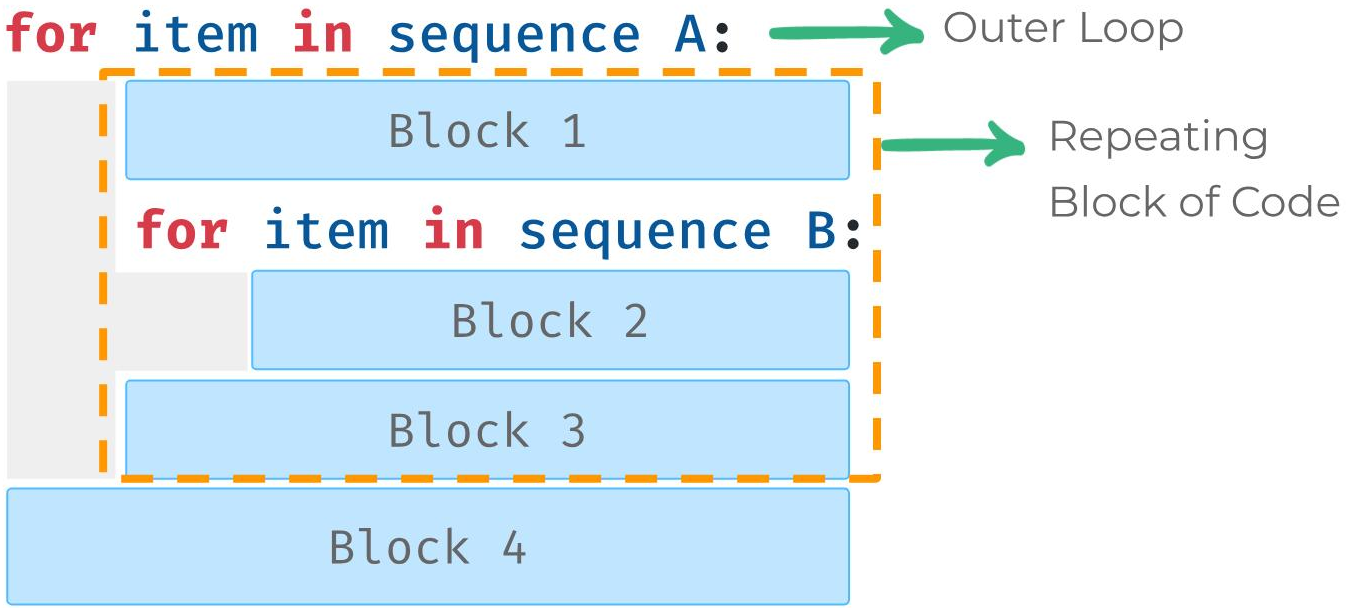
Nested Loops



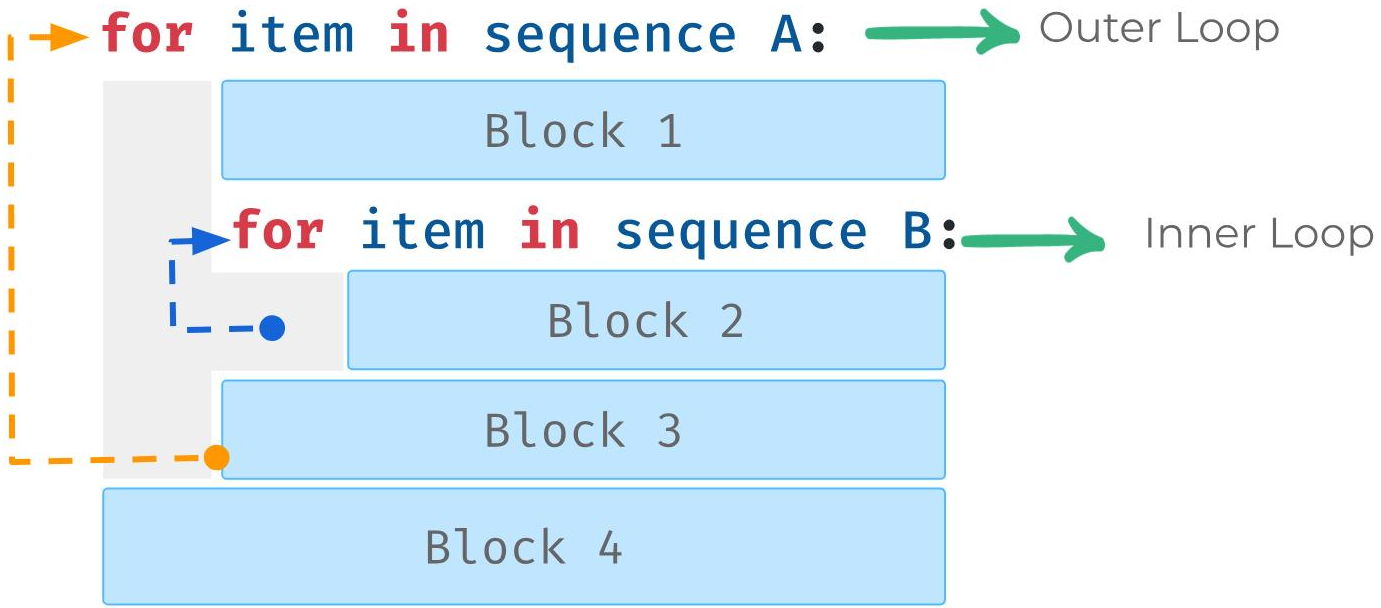
Nested Loops



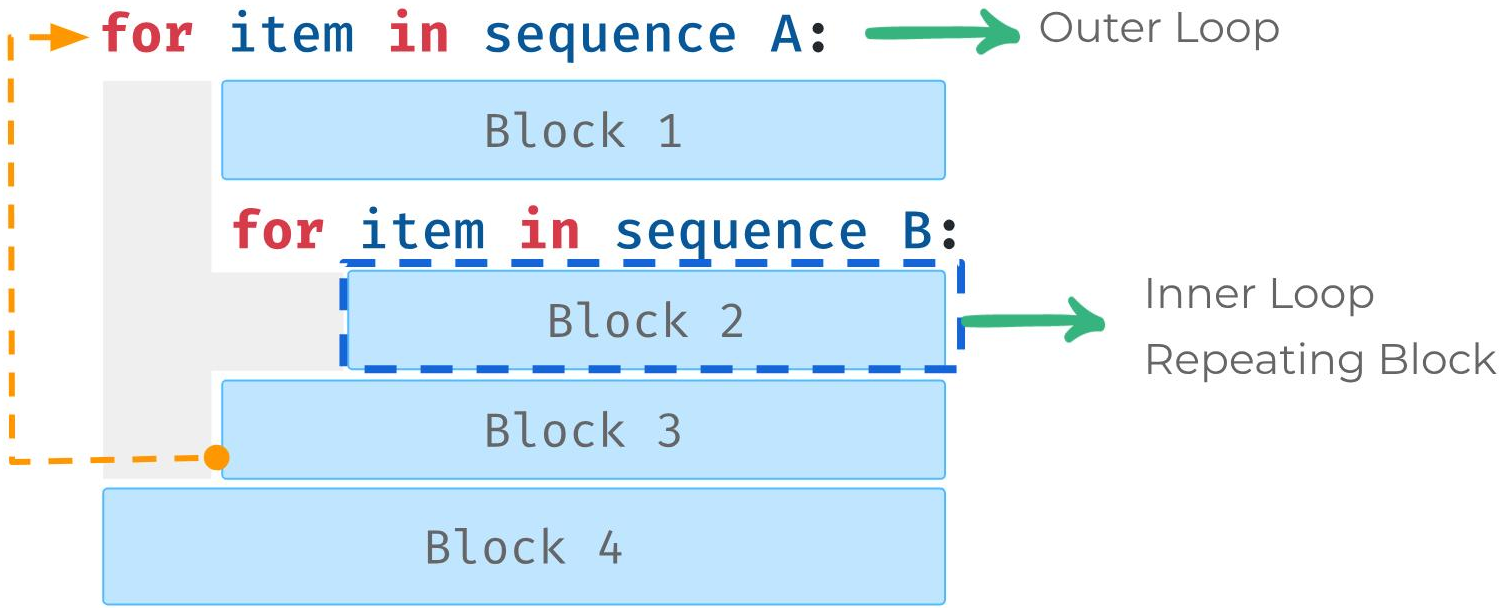
Nested Loops



Nested Loops



Nested Loops



Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

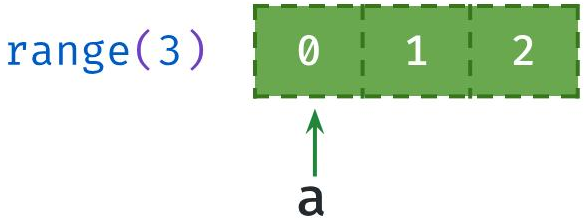
Lets see the

step by step execution

Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```



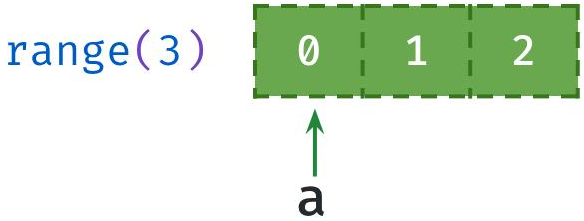
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
```



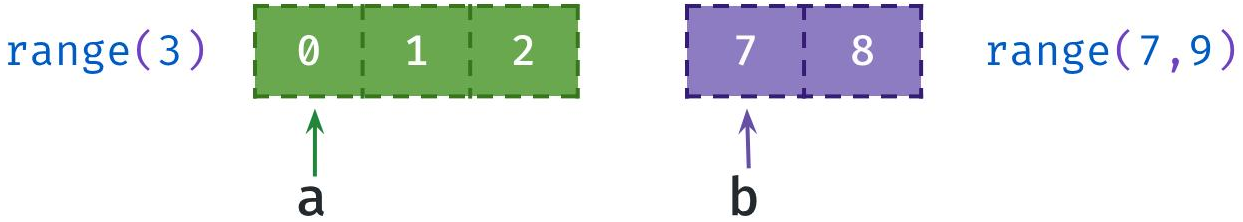
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
```



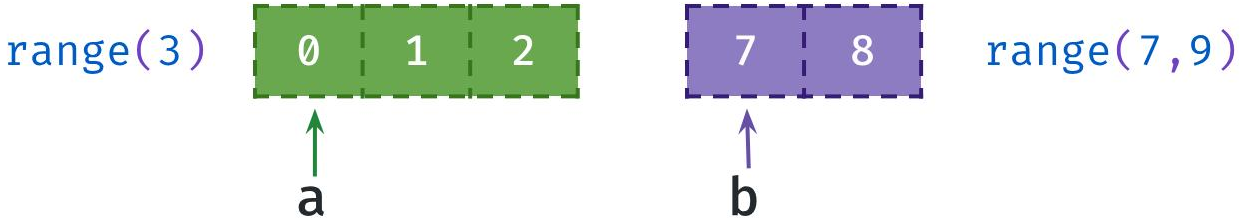
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
  inner : 7
```



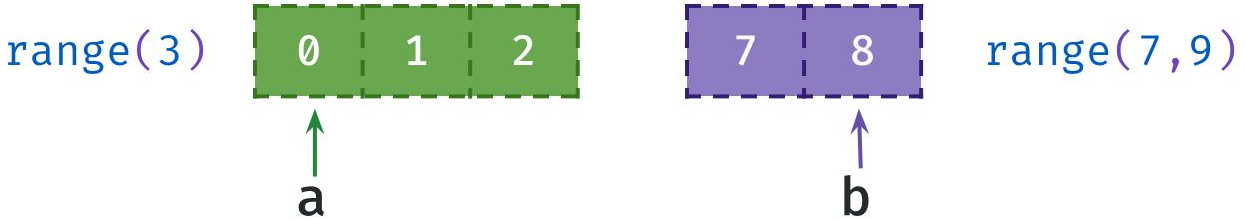
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
  inner : 7
```



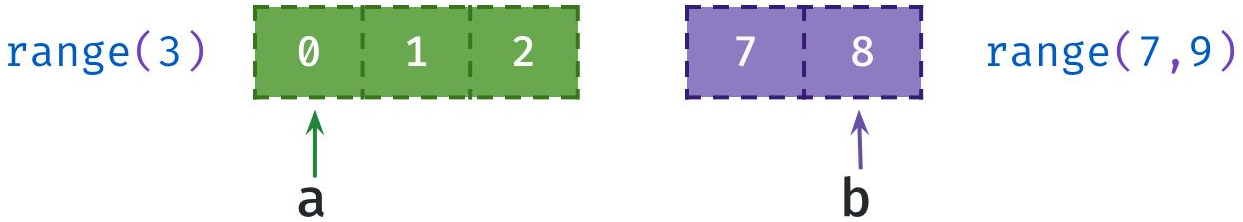
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
  inner : 7
  inner : 8
```



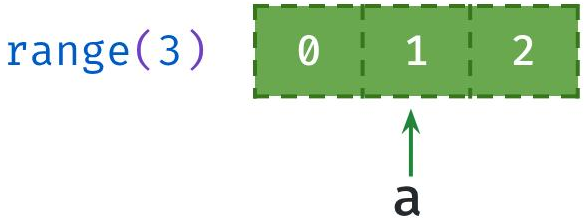
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
  inner : 7
  inner : 8
```



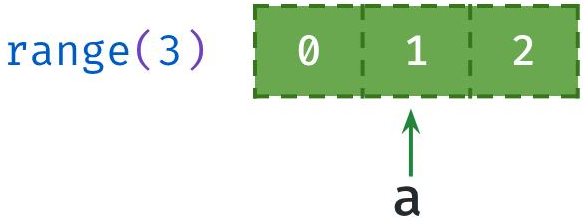
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
  inner : 7
  inner : 8
outer : 1
```



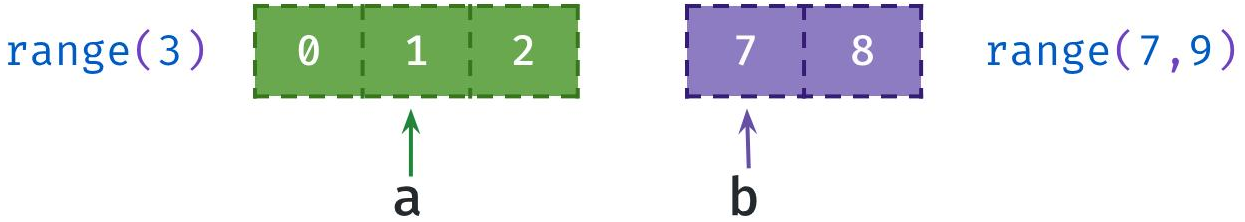
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
  inner : 7
  inner : 8
outer : 1
```



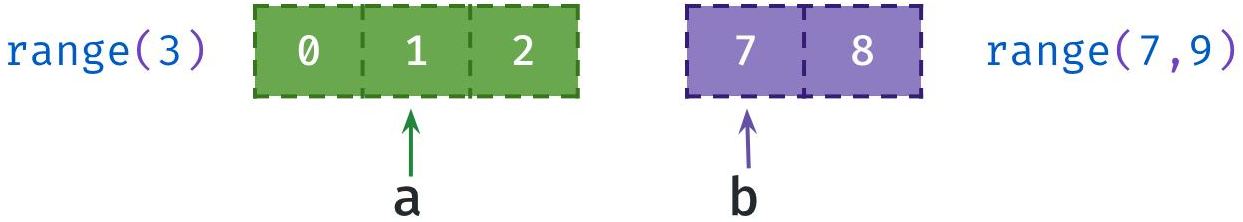
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
  inner : 7
  inner : 8
outer : 1
  inner : 7
```



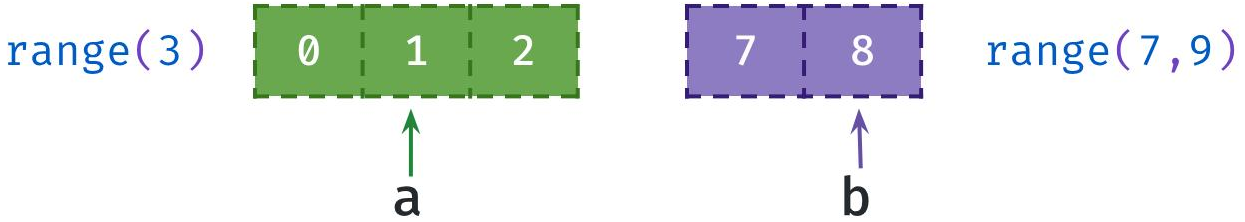
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
  inner : 7
  inner : 8
outer : 1
  inner : 7
```



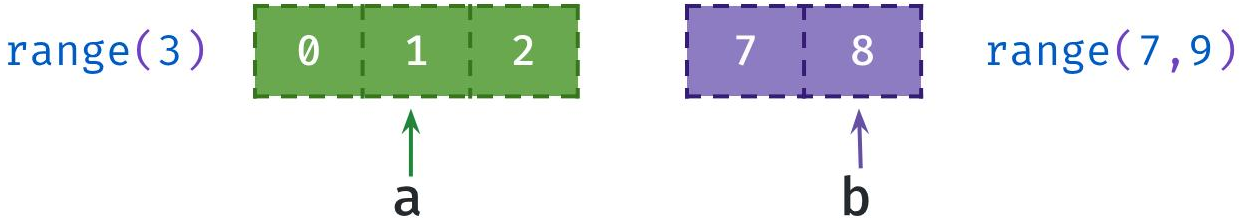
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
  inner : 7
  inner : 8
outer : 1
  inner : 7
  inner : 8
```



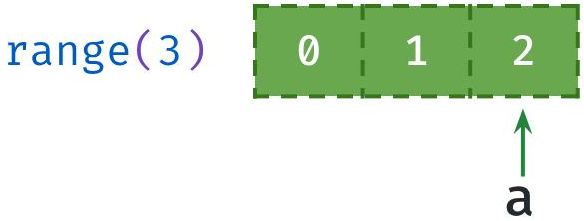
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
  inner : 7
  inner : 8
outer : 1
  inner : 7
  inner : 8
```



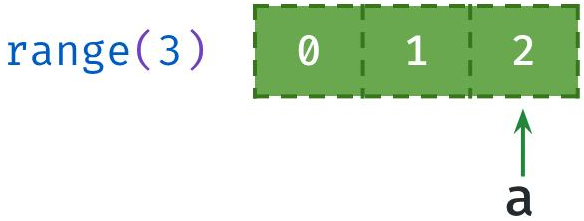
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
  inner : 7
  inner : 8
outer : 1
  inner : 7
  inner : 8
outer : 2
```



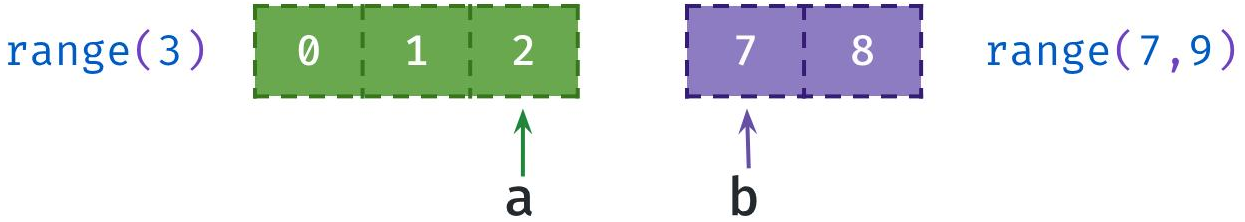
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
  inner : 7
  inner : 8
outer : 1
  inner : 7
  inner : 8
outer : 2
```



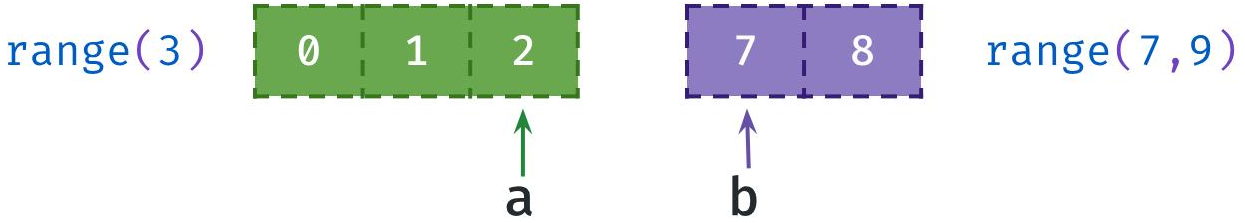
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
  inner : 7
  inner : 8
outer : 1
  inner : 7
  inner : 8
outer : 2
  inner : 7
```



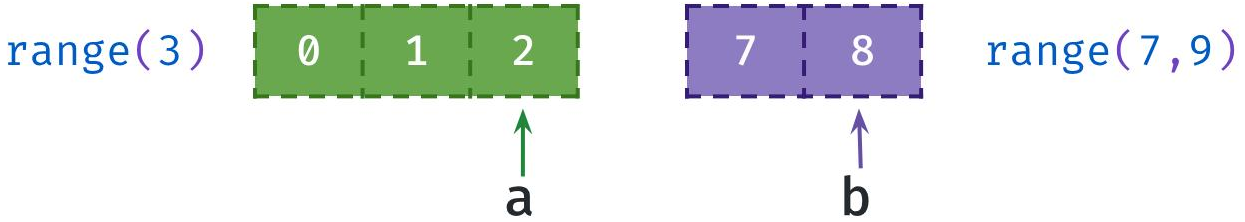
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
  inner : 7
  inner : 8
outer : 1
  inner : 7
  inner : 8
outer : 2
  inner : 7
```



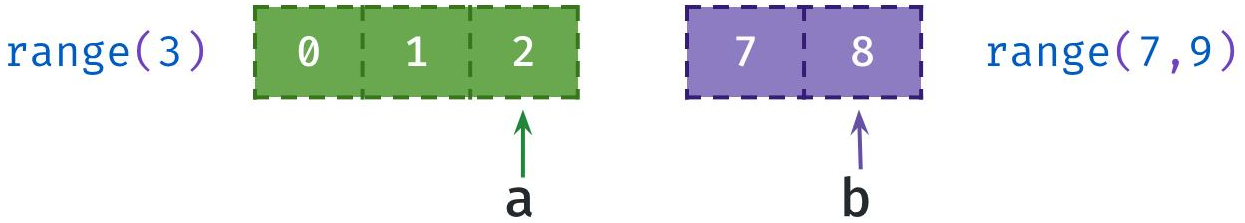
Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
  inner : 7
  inner : 8
outer : 1
  inner : 7
  inner : 8
outer : 2
  inner : 7
  inner : 8
```



Nested Loops

Code

```
for a in range(3):
    print("outer : " + str(a))
    for b in range(7,9):
        print("  inner : " + str(b))
print("End")
```

Output

```
outer : 0
  inner : 7
  inner : 8
outer : 1
  inner : 7
  inner : 8
outer : 2
  inner : 7
  inner : 8
End
```

Question

Triangular Pattern with Numbers

Write a program to print
a **triangular pattern**
with numbers in N rows,
where integer N is given as input

Input

```
4
```

Output

```
1
1 2
1 2 3
1 2 3 4
```

NXT WAVE

Question

Input / Output Format

Input Format

First line will contain a positive integer





NXT WAVE

Question

Input / Output Format

Output Format

Print a triangular pattern with numbers

Note: There is a space after every ***number***

Input

```
5
```

Output

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

Possible Mistakes

Wrong Iteration

Code

```
num = int(input())
for row_num in range(1, num):
    row_out = ""
    for seq_num in range(1, row_num + 1):
        row_out = row_out + str(seq_num) + " "
    print(row_out)
```

Input

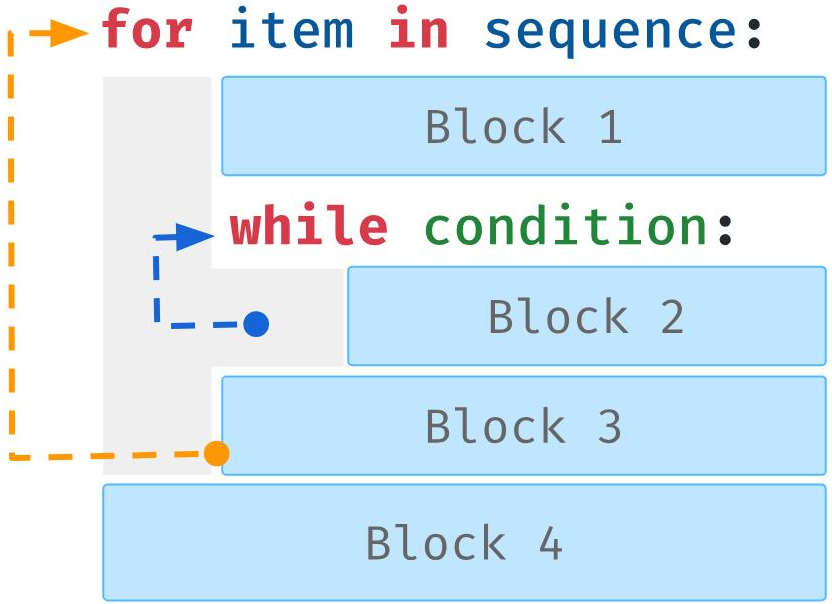
```
3
```

Output

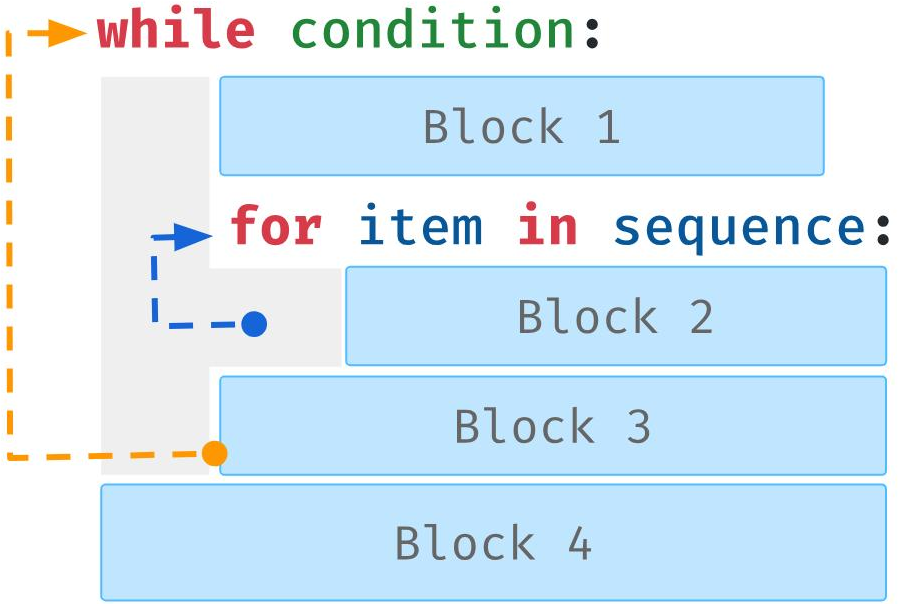
```
1
1 2
```

NXT WAVE

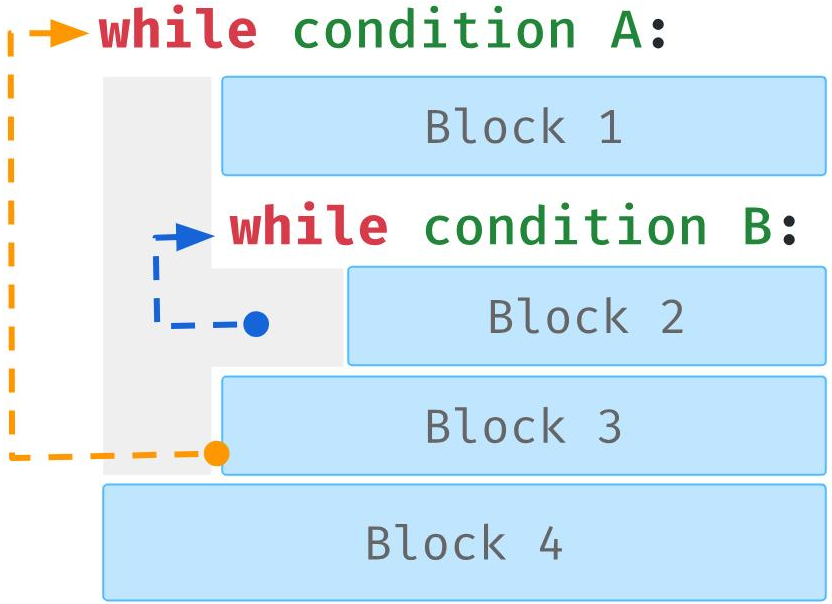
Nested Loops



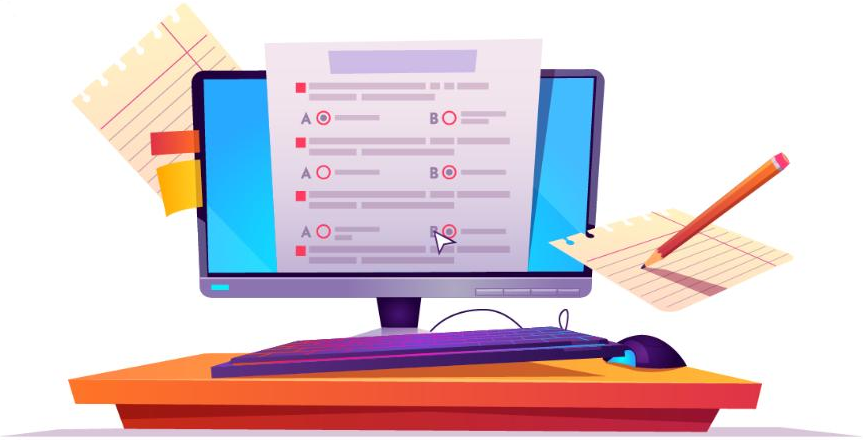
Nested Loops



Nested Loops



Practice



Daily Challenge

Identify The Mistake

Code

```
n = int(input())
for row_num in range(0,n+1):
    row_output = ""
    seq_num = row_num
    while seq_num > 0:
        row_output = row_output + str(seq_num)
        seq_num = seq_num - 1
    print(row_output)
```

Input

```
3
```

Expected Output

```
321
21
1
```

NXT WAVE

NXT WAVE

THANK YOU



ALL THE BEST